Chalana Sangeeth

Embilipitiya | dcsangeeth@gmail.com | 071 9130003 | My Portfolio | dcsangeeth | dcsangeeth

Professional Summery

AI/ML Engineer with hands-on experience developing **AI-powered automation**, **Agentic AI workflows**, **Retrieval-Augmented Generation (RAG)** systems, and **Flask REST APIs** during internships at **Sri Lanka Telecom**. Strong foundation in **Python**, **Machine Learning**, **Deep Learning**, and **Generative AI**, with additional experience in **seismic signal processing**, **ESP32-based embedded AI**, and **IoT system development**. **BSc.(Hons) Engineering** degree requirements completed in **Electrical and Electronic Engineering** at **South Eastern University of Sri Lanka**.
Available for full-time opportunities in **AI/ML Engineering** and **AI/ML + IoT Engineering**.

Education

BSc(Hons) in Electrical & Electronic Engineering *Aug 2022 – Oct 2026*
South Eastern University of Sri Lanka

- GPA: 3.35/4.0

Experience

Artificial Intelligence(AI) Engineer (Intern) *Colombo, SL*
Sri Lanka Telecom *Aug 2025 – Oct 2025*

- Developed an **AI-driven back-office automation system** using **LangGraph, Gemini, RAG, and n8n** for automated request processing and information retrieval.
- Built **agentic AI workflows and Flask REST APIs** for usage analysis and document-based problem resolution, with **LangSmith** for workflow monitoring and debugging.

Software Engineer (Intern) *Colombo, SL*
Sri Lanka Telecom *May 2024 – Aug 2024*

- Developed and enhanced **PowerProx**, an internal asset-management system for tracking **rectifiers, generators, batteries, SPDs, and UPS systems**.
- Built **Flutter/Dart features and REST API integrations**, digitizing asset management and reducing manual record handling.

Projects

Geophone-Based Seismic Sensing Approach for Elephant Detection and Classification *Jan 2026 - Oct 2026*

- Developing a non-invasive elephant detection system using geophone-based seismic signals and digital signal processing.
- Implementing **signal preprocessing, feature extraction, and Machine Learning classification** to distinguish elephant-related seismic activity from other sources.
- Integrating ESP32-S3 embedded processing for real-time signal analysis and **AI-based prediction** .
- Working with sensor data acquisition and IoT components for field-oriented monitoring.

Industrial Engineering Project – PowerProx (SLT) May 2024 - Aug 2024

- Developed **Flutter/Dart** features and integrated **REST APIs** for real-time asset tracking and updates.
- Integrated **Gemini API** for **AI-assisted quotation retrieval and processing**.
- **Tools:** Flutter, Dart, REST APIs, Gemini API, **Generative AI**

Smart Route Optimization System *May 2025 -May 2025*

- Developed a **Flask-based** route optimization system using **Dijkstra's algorithm** and **Google Maps API**.

Accommodation Booking Mobile App *Apr 2025 - Jan 2026*

- Developed a **Flutter** application with **Firebase authentication** and image uploads.
- Built **Flask REST APIs** with **MongoDB** for property management.

Certifications

Advanced Learning Algorithms *Jun - 2026*

- DeepLearning.AI on Coursera

AI / ML Engineer - Stage 1 *Apr - 2025*

- Centre of Open and Distance Education - SLIIT

AI / ML Engineer - Stage 2 *Apr - 2025*

- Centre of Open and Distance Education - SLIIT

Supervised Machine Learning - Regression and Classification *Jul - 2024*

- DeepLearning.AI on Coursera

AgniXtream Hackathon Competition *Dec - 2024*

- 1 Runners Up at 6 hours Hackathon
- IEEE Sri Lanka Section

Technical Skills

Programming: Python, C++, Dart

AI / Machine Learning: Machine Learning, Deep Learning, TensorFlow, scikit-learn, NumPy, Pandas

Generative & Agentic AI: RAG, LangChain, LangGraph, Gemini API, Agentic AI, n8n, LangSmith

Signal Processing: Digital Signal Processing, Seismic Signal Processing, Filtering, FFT, Feature Extraction

Embedded & IoT: ESP32, ESP32-S3, Embedded AI, Sensors, ADC, IoT Systems, Real-Time Data Acquisition

Backend & Software: Flask, REST APIs, MongoDB, Firebase, Flutter, Git

Tools: MATLAB

References

Prof. Dr. Eng. Thushara Weerawardane
Professor of Engineering
Dean , Faculty of Engineering
General Sir John Kotelawala Defence University (KDU)
Phone: +94 77 444 1419
Email: tlw@kdu.ac.lk

Prof. M. A. L. A. Haleem
Professor of Electrical and Electronic Engineering
Dean, Faculty of Engineering
South Eastern University of Sri Lanka (SEUSL)
University Park, Oluvil, #32360, Sri Lanka
Phone: +94 76 724 1063
Email: malahaleem@seu.ac.lk

Eng. Dinesh Piyasamara
Research Executive (AI & ML Solutions)
AI/ML Engineer
MSc in FinMath (R) – UOM
BSc (Hons) Engineering – UOJ
Mobile**:** +94 71 1234 143
Email: dineshpi@slt.com.lk

Eng. Oshadha Ranaweera
Chartered Engineer — MSc. Eng. — BSc. Eng. (Hons)
AMIE(SL) — MIET(UK)
Power Operation Engineer
AC and Power Operation Section
Sri Lanka Telecom
Phone: +94 71 864 9622
Email: oshadha@slt.com.lk